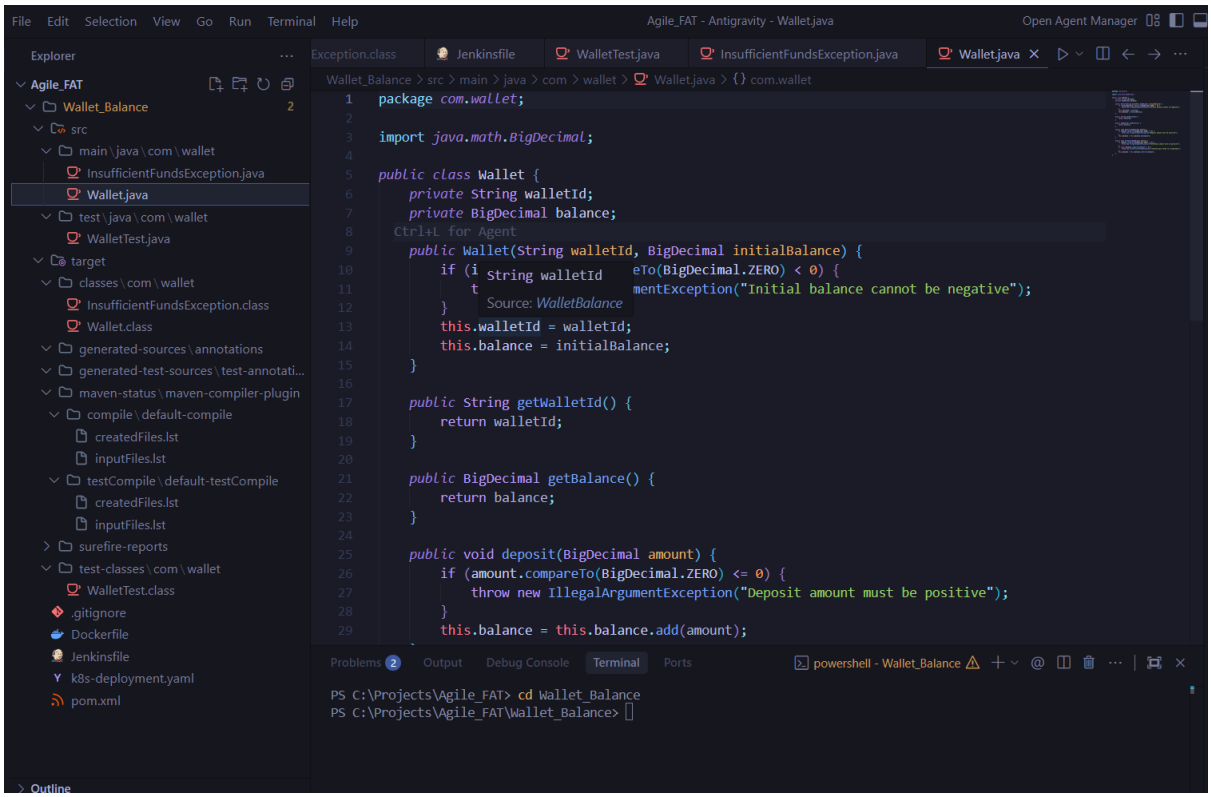
23MIS0325

Pravin G

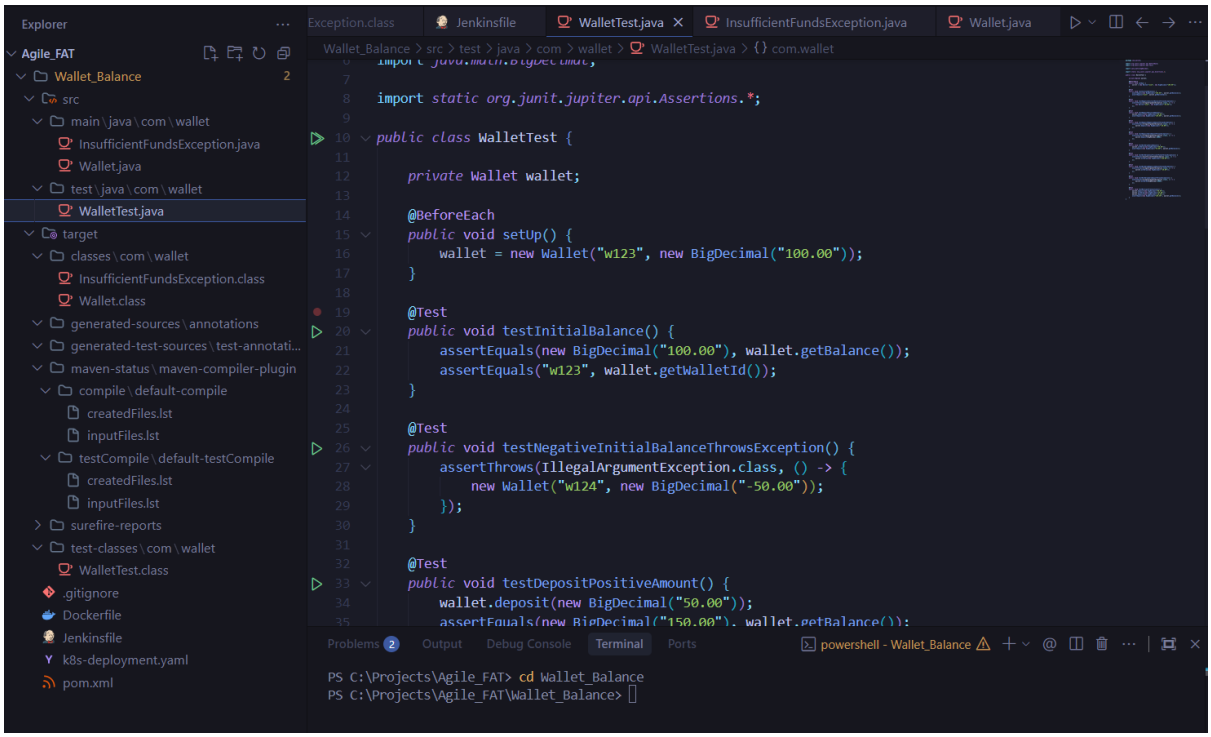
Agile FAT

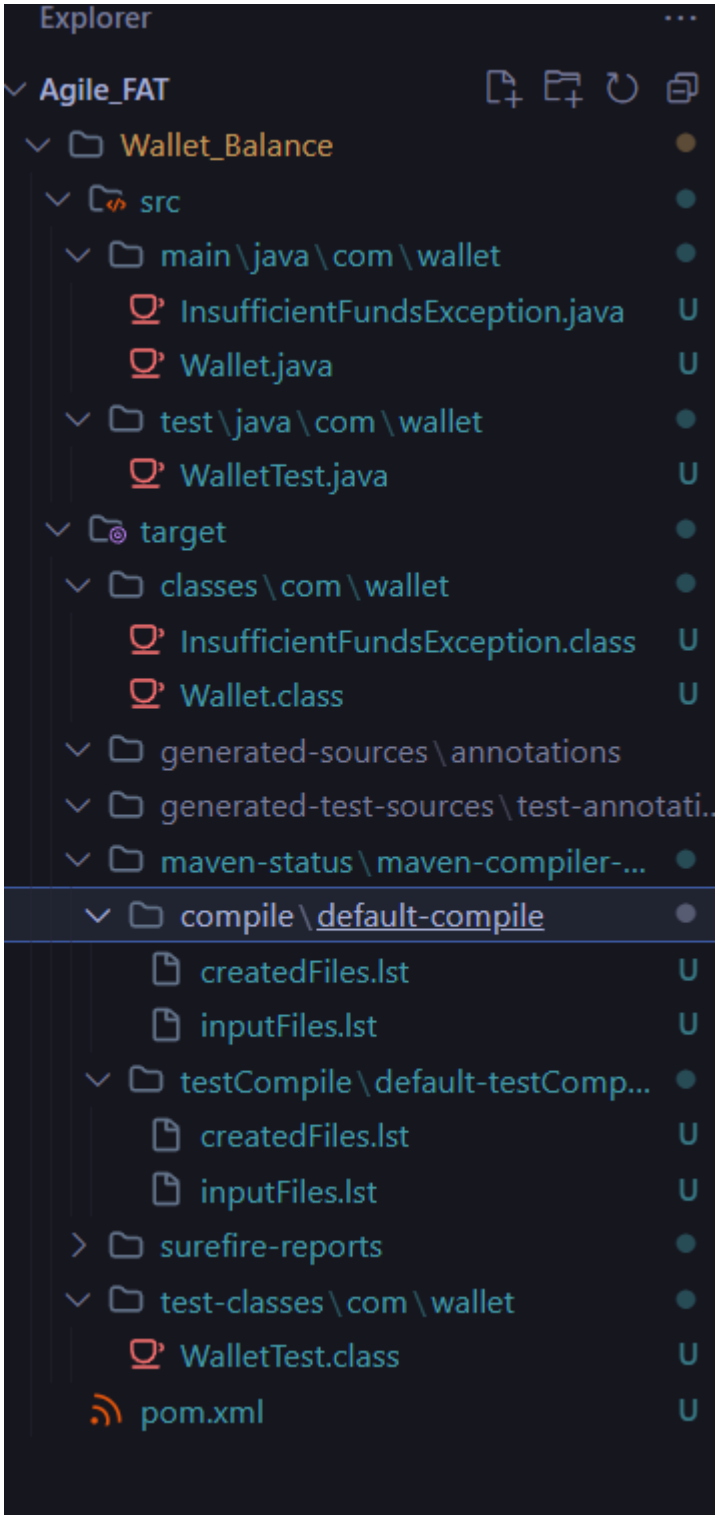
Github url: https://github.com/Pravin98651/Wallet_Balance

Java Program Development Code:



Junit Test case code




Explorer
Agile_FAT
Wallet_Balance
src
main\java\com\wallet
InsufficientFundsException.java
Wallet.java
test\java\com\wallet
WalletTest.java
target
classes\com\wallet
InsufficientFundsException.class
Wallet.class
generated-sources\annotations
generated-test-sources\test-annotati...
maven-status\maven-compiler-...
compile\default-compile
createdFiles.lst
inputFiles.lst
testCompile\default-testComp...
createdFiles.lst
inputFiles.lst
surefire-reports
test-classes\com\wallet
WalletTest.class
pom.xml

Pom.xml file

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/mave
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.wallet</groupId>
    <artifactId>WalletBalance</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>11</maven.compiler.source>
        <maven.compiler.target>11</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <junit.version>5.9.2</junit.version>
    </properties>

    <dependencies>
        <!-- JUnit 5 -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-engine</artifactId>
            <version>${junit.version}</version>
```

```
PS C:\Projects\Agile_FAT> cd Wallet_Balance
PS C:\Projects\Agile_FAT\Wallet_Balance>
```

Docker File

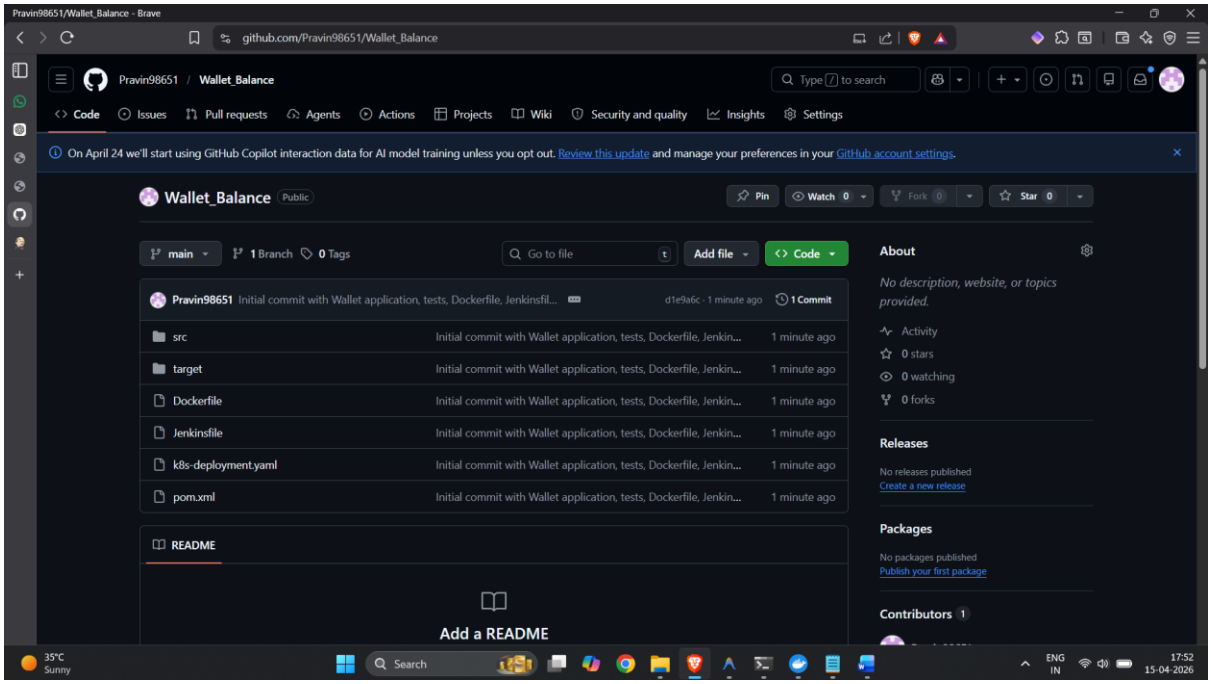
```
FROM openjdk:11-jre-slim

WORKDIR /app

COPY target/WalletBalance-1.0-SNAPSHOT.jar app.jar

ENTRYPOINT ["java", "-jar", "app.jar"]
```

k8s-deployment.yaml | Jenkinsfile | WalletTest.java | InsufficientFundsException.java | Wallet.java

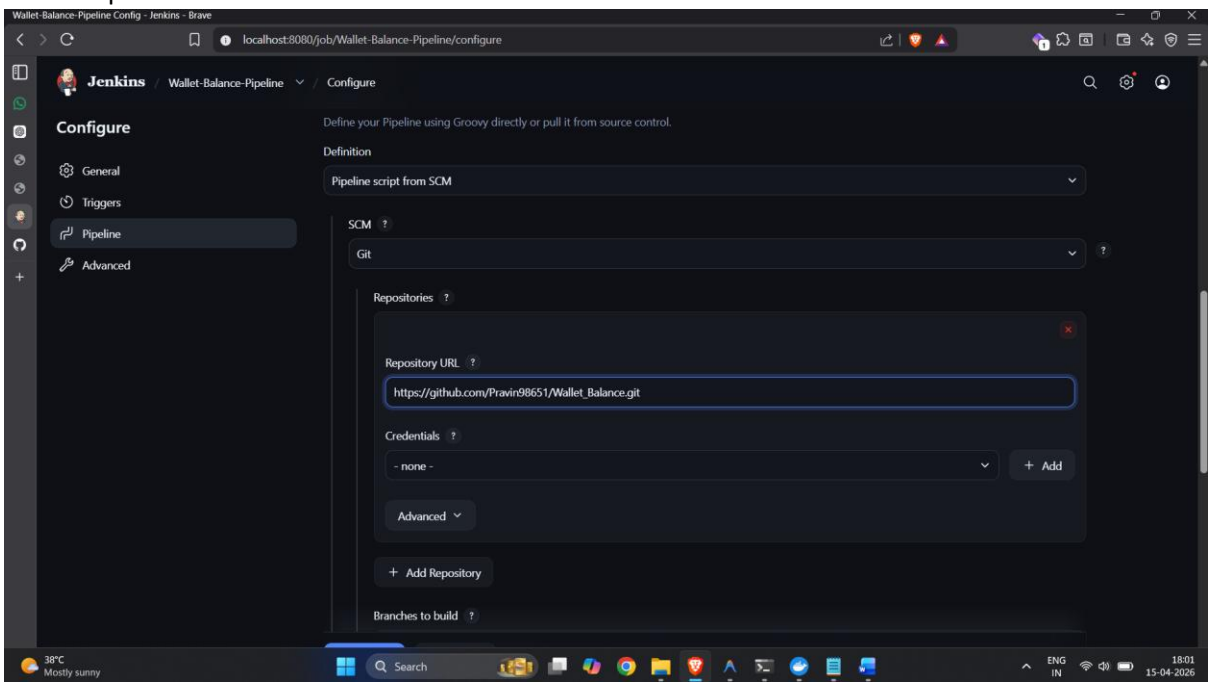
Wallet_Balance > k8s-deployment.yaml

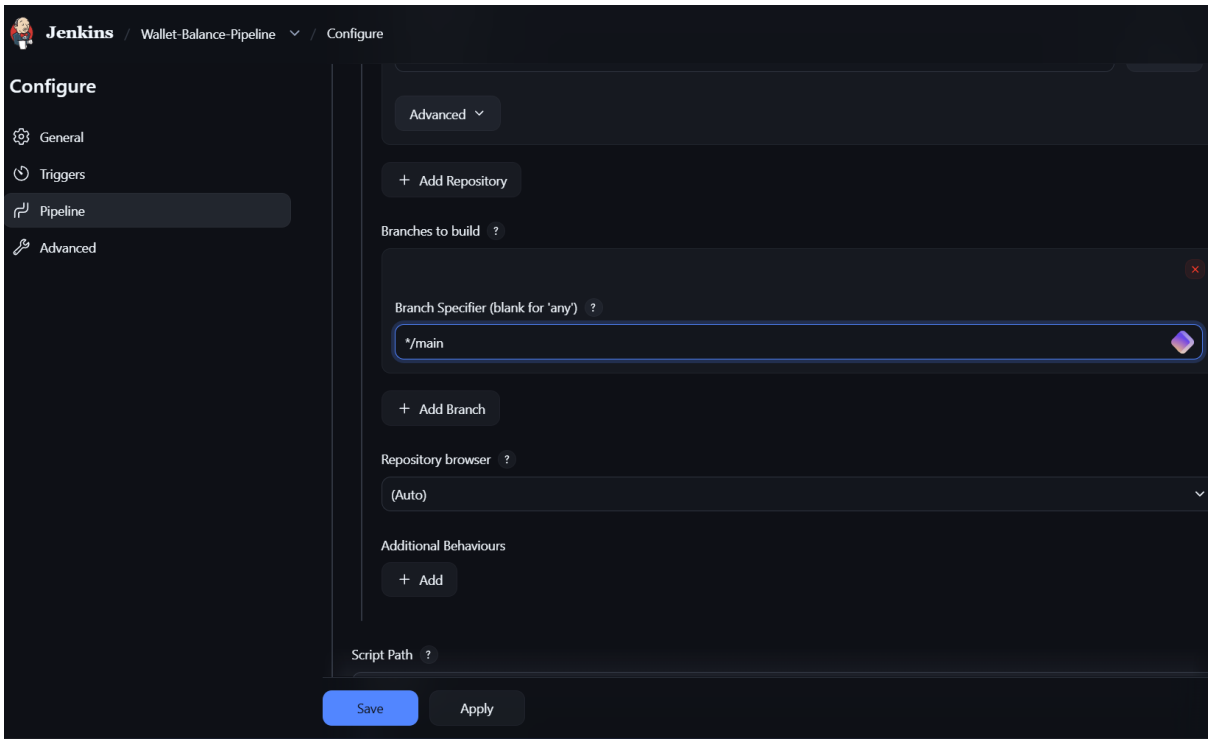
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wallet-balance-app
  labels:
    app: wallet-balance
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wallet-balance
  template:
    metadata:
      labels:
        app: wallet-balance
    spec:
      containers:
      - name: wallet-balance
        image: pravin98651/wallet-balance:latest
        ports:
        - containerPort: 8080
        resources:
          requests:
            memory: "64Mi"
            cpu: "250m"
          limits:
            memory: "128Mi"
            cpu: "500m"
```

Github pushed

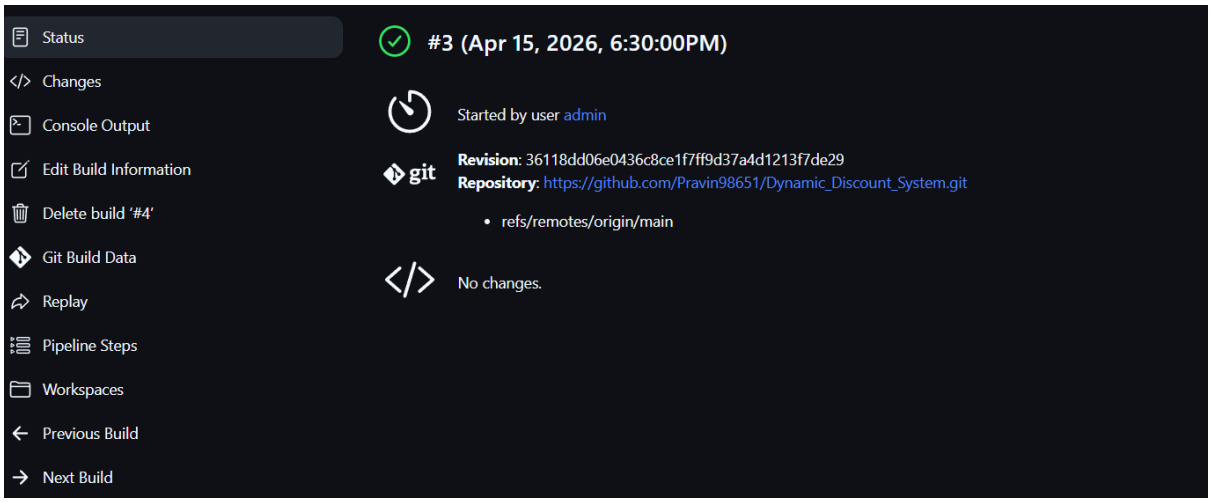


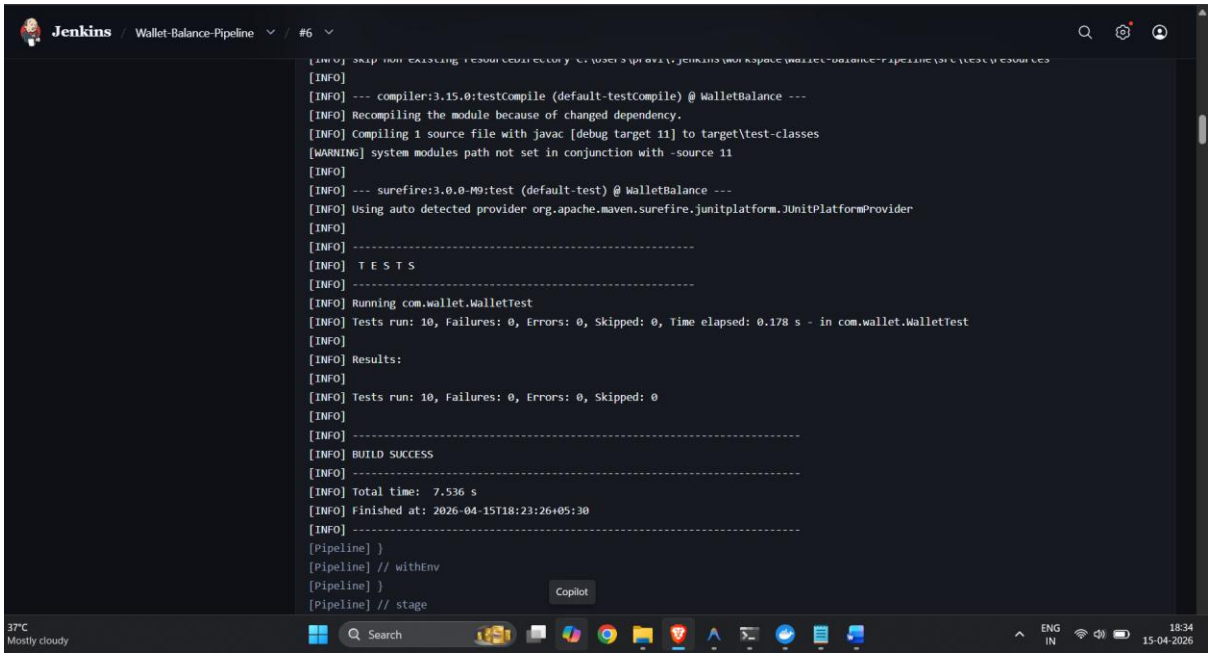
Jenkin Pipeline Creation

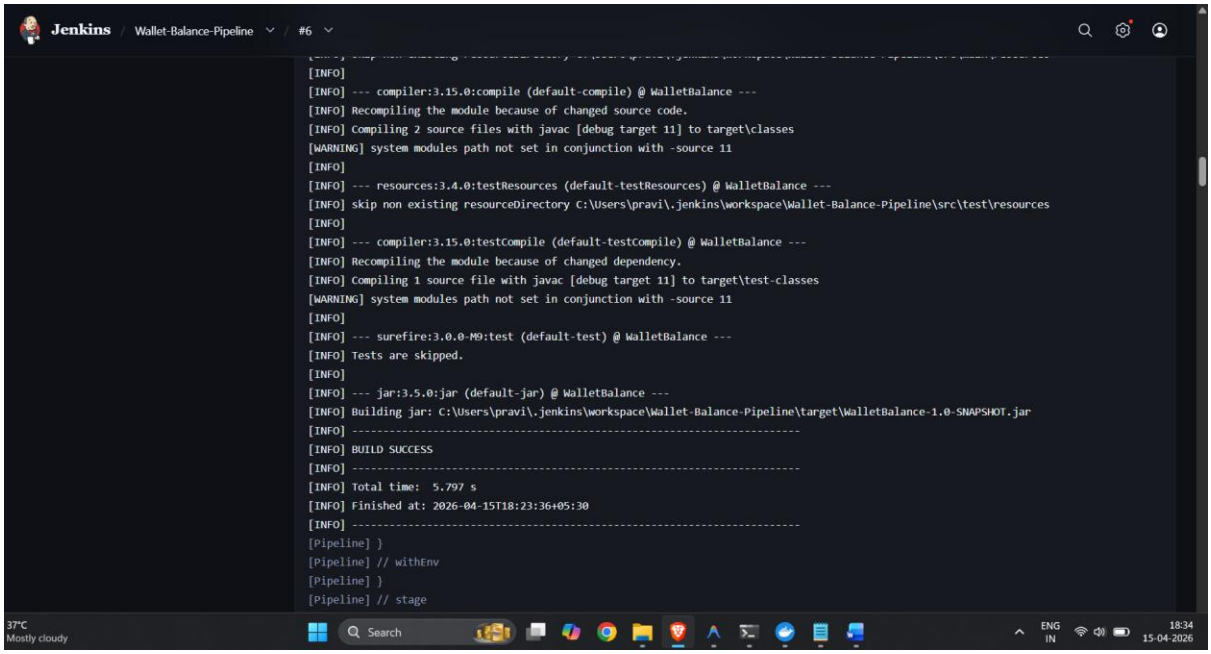




Jenkin Pipeline Creation Successful with junit testing and deployment in docker and Kubernetes







Junit test case run

```
PS C:\Projects\Agile_FAT\Wallet_Balance> mvn test
>>
[INFO] ----------------------< com.wallet:WalletBalance >----------------------
[INFO] Building WalletBalance 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ WalletBalance ---
[INFO] skip non existing resourceDirectory C:\Projects\Agile_FAT\Wallet_Balance\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ WalletBalance ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ WalletBalance ---
[INFO] skip non existing resourceDirectory C:\Projects\Agile_FAT\Wallet_Balance\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ WalletBalance ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.0.0-M9:test (default-test) @ WalletBalance ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.wallet.WalletTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.197 s - in com.wallet.Wallet
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.954 s
```

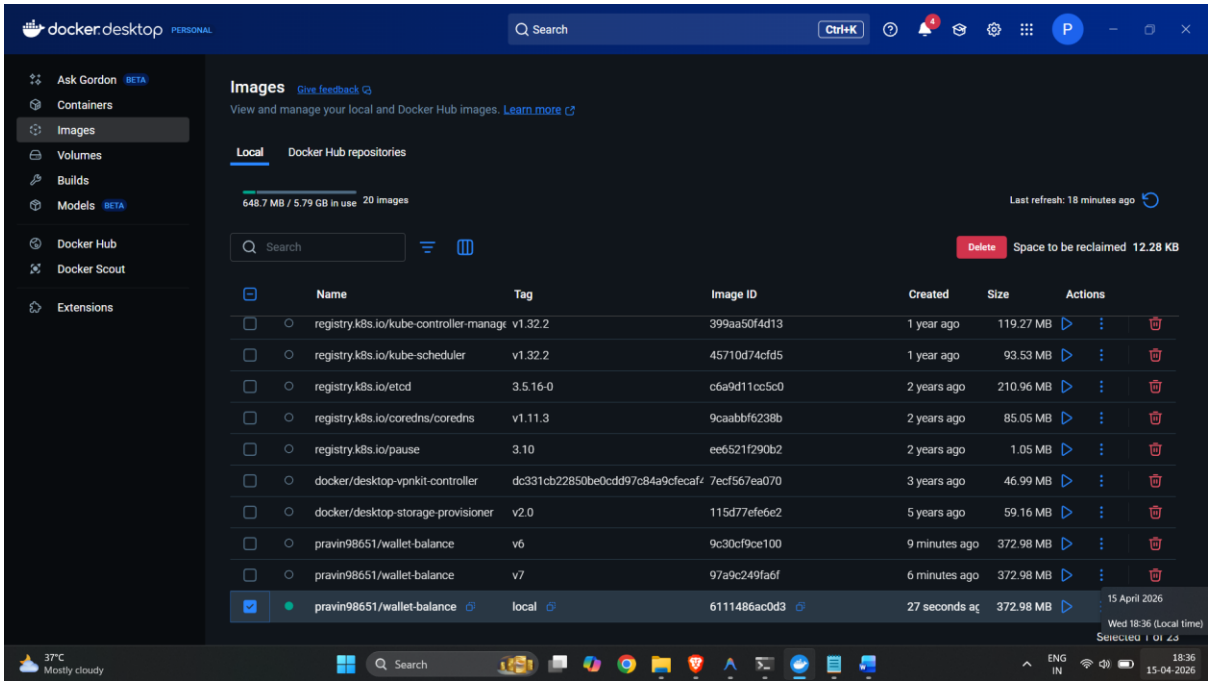
Docker images

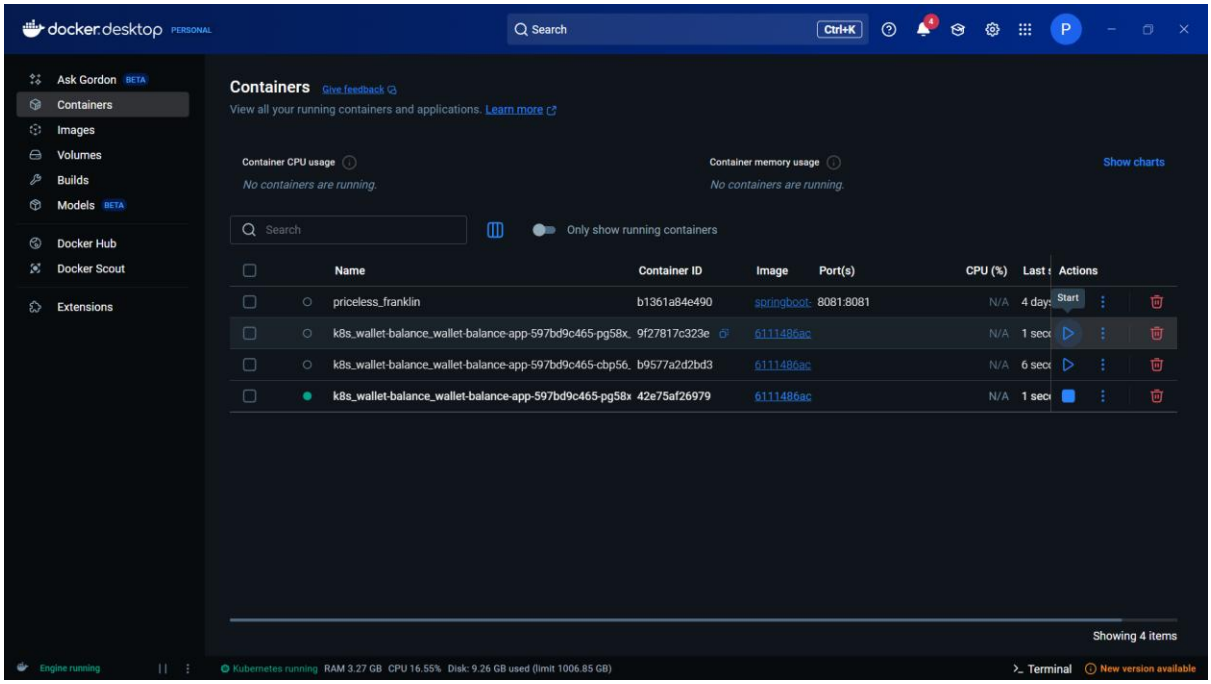
```
Problems 2   Output   Debug Console   Terminal   Ports                powershell - Wallet_Bal

PS C:\Projects\Agile_FAT\Wallet_Balance> docker images
>>
REPOSITORY                                  TAG
 IMAGE ID       CREATED          SIZE
pravin98651/wallet-balance                  local
 6111486ac0d3   3 minutes ago    373MB
pravin98651/wallet-balance                  v7
 97a9c249fa6f   8 minutes ago    373MB
pravin98651/wallet-balance                  v6
 9c30cf9ce100   12 minutes ago   373MB
leave-system                                v3
 312fd7d42c93   3 days ago       674MB
```

```
PS C:\Projects\Agile_FAT\Wallet_Balance> kubectl get pods
>>
NAME                                  READY   STATUS             RESTARTS      AGE
wallet-balance-app-597bd9c465-cbp56   0/1     CrashLoopBackOff   5 (74s ago)   4m13s
wallet-balance-app-597bd9c465-pg58x   0/1     CrashLoopBackOff   5 (70s ago)   4m13s
PS C:\Projects\Agile_FAT\Wallet_Balance>
```

```
PS C:\Projects\Agile_FAT\Wallet_Balance> kubectl get services
>>
NAME                    TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes              ClusterIP      10.96.0.1       <none>        443/TCP        3d4h
wallet-balance-service  LoadBalancer   10.108.50.105   localhost     80:31182/TCP   4m26s
PS C:\Projects\Agile_FAT\Wallet_Balance>
```

```
PS C:\Projects\Agile_FAT\Wallet_Balance> kubectl get deployments
>>
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
wallet-balance-app    0/2     2            0           4m39s
PS C:\Projects\Agile_FAT\Wallet_Balance>
```


docker desktop PERSONAL
Search Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Builds
Models BETA
Docker Hub
Docker Scout
Extensions
Images Give feedback
View and manage your local and Docker Hub images. Learn more
Local
Docker Hub repositories
648.7 MB / 5.79 GB in use 20 images
Last refresh: 18 minutes ago
Search
Delete
Space to be reclaimed 12.28 KB
Name Tag Image ID Created Size Actions
registry.k8s.io/kube-controller-manage v1.32.2 399aa50f4d13 1 year ago 119.27 MB
registry.k8s.io/kube-scheduler v1.32.2 45710d74cfd5 1 year ago 93.53 MB
registry.k8s.io/etcd 3.5.16-0 c6a9d11cc5c0 2 years ago 210.96 MB
registry.k8s.io/coredns/coredns v1.11.3 9caabbf6238b 2 years ago 85.05 MB
registry.k8s.io/pause 3.10 ee6521f290b2 2 years ago 1.05 MB
docker/desktop-vpnkit-controller dc331cb22850be0cdd97c84a9cfecaf4 7ecf567ea070 3 years ago 46.99 MB
docker/desktop-storage-provisioner v2.0 115d77efe6e2 5 years ago 59.16 MB
pravin98651/wallet-balance v6 9c30cf9ce100 9 minutes ago 372.98 MB
pravin98651/wallet-balance v7 97a9c249fa6f 6 minutes ago 372.98 MB
pravin98651/wallet-balance local 6111486ac0d3 27 seconds ag 372.98 MB
15 April 2026
Wed 18:36 (Local time)
37°C
Mostly cloudy
ENG
IN
18:36
15-04-2026


docker desktop PERSONAL
Search Ctrl+K
Ask Gordon BETA
Containers
Images
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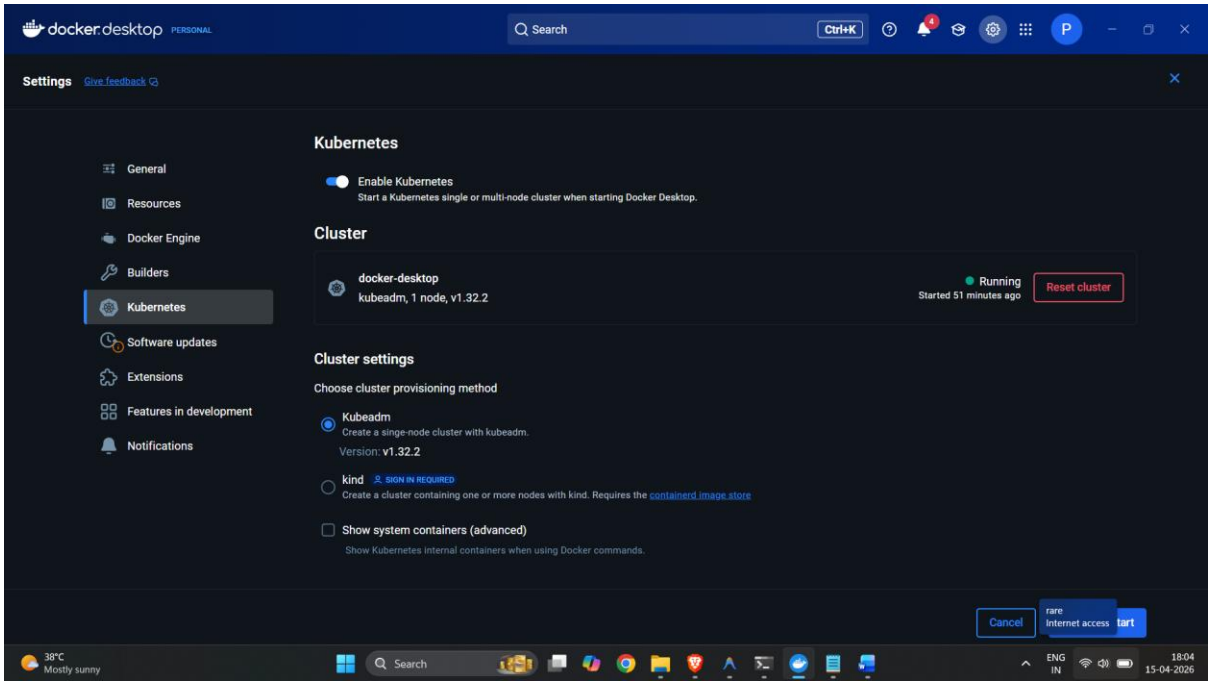
Builds
Models BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
View all your running containers and applications. Learn more
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name Container ID Image Port(s) CPU (%) Last Actions
priceless_franklin b1361a84e490 springboot- 8081:8081 N/A 4 days
Start
k8s_wallet-balance_wallet-balance-app-597bd9c465-pg58x_ 9f27817c323e 6111486ac N/A 1 seco
k8s_wallet-balance_wallet-balance-app-597bd9c465-cbp56_ b9577a2d2bd3 6111486ac N/A 6 seco
k8s_wallet-balance_wallet-balance-app-597bd9c465-pg58x 42e75af26979 6111486ac N/A 1 seco
Showing 4 items
Engine running
Kubernetes running RAM 3.27 GB CPU 16.55% Disk: 9.26 GB used (limit 1006.85 GB)
Terminal
New version available


docker desktop PERSONAL
Search
Ctrl+K
Settings
Give feedback
General
Resources
Docker Engine
Builders
Kubernetes
Software updates
Extensions
Features in development
Notifications
Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
Cluster
docker-desktop
kubeadm, 1 node, v1.32.2
Running
Started 51 minutes ago
Reset cluster
Cluster settings
Choose cluster provisioning method
Kubeadm
Create a singe-node cluster with kubeadm.
Version: v1.32.2
kind
SIGN IN REQUIRED
Create a cluster containing one or more nodes with kind. Requires the containerd image store
Show system containers (advanced)
Show Kubernetes internal containers when using Docker commands.
Cancel
rare
Internet access
tart
38°C
Mostly sunny
Search
ENG
IN
18:04
15-04-2026